Dietary intakes of flavonols and flavones and coronary heart disease in US women.

Dietary flavonols and flavones are subgroups of flavonoids that have been suggested to decrease the risk of coronary heart disease (CHD). The authors prospectively evaluated intakes of flavonols and flavones in relation to risk of nonfatal myocardial infarction and fatal CHD in the Nurses' Health Study. They assessed dietary information from the study's 1990, 1994, and 1998 food frequency questionnaires and computed cumulative average intakes of flavonols and flavones. Cox proportional hazards regression with time-varying variables was used for analysis. During 12 years of follow-up (1990-2002), the authors documented 938 nonfatal myocardial infarctions and 324 CHD deaths among 66,360 women. They observed no association between flavonol or flavone intake and risk of nonfatal myocardial infarction or fatal CHD. However, a weak risk reduction for CHD death was found among women with a higher intake of kaempferol, an individual flavonol found primarily in broccoli and tea. Women in the highest quintile of kaempferol intake relative to those in the lowest had a multivariate relative risk of 0.66 (95% confidence interval: 0.48, 0.93; p for trend = 0.04). The lower risk associated with kaempferol intake was probably attributable to broccoli consumption. These prospective data do not support an inverse association between flavonol or flavone intake and CHD risk.